

22/01/25

Lecture 6

General Prescription to exactly solve 1D Ising Models & other generalizations



a) Identify discrete blocks of interacting spins. For nearest neighbour each block contains only one spin. For two neighbour interactions, each block has two spins & 4 states, & so on.

b) Let $H(S_i, S_j)$ be the energy of interaction b/w two nearby blocks. Associate a matrix T (Transfer matrix) for the interaction b/w two blocks.

$$e^{H(S_i, S_j)} = \langle S_i | \hat{T} | S_j \rangle$$

c) Partition function is given by

$$Z = \text{Trace} (T^N) \quad (N \rightarrow \text{blocks})$$

$$Z = \sum_m \lambda_m^N \quad (\lambda_m \Rightarrow \text{eigen values of } T)$$

- d) Use Perron-Frobenius Theorem,
"Any real square matrix with all +ve entries has a unique eigen value of largest magnitude & that eigen value is +ve, & its corresponding eigen vector has all +ve entries."

$$\text{Thus } \lambda_0 > \lambda_k \quad (k \neq 0)$$

$$\Rightarrow Z = \lambda_0^N \left[1 + \sum_{m \neq 0} \left(\frac{\lambda_m}{\lambda_0} \right)^N \right]$$

In thermodynamic limit $N \rightarrow \infty$, so

$$\left(\frac{\lambda_m}{\lambda_0} \right)^N \rightarrow 0, \text{ hence } \underline{\underline{Z \sim \lambda_0^N}}$$

- e) Find all required thermodynamic quantities from Z (Partition function)
- f) Find the spin-spin correlation function
 $\langle S_i S_{i+r} \rangle = \text{Trace} (S_i S_{i+r} e^{-\beta \hat{H}})$

use the Transfer matrix method.

- g) Find $\sum_r G(r) = \frac{\chi_s}{\beta} = \sum_r \langle S_i S_{i+r} \rangle$
& verify with $\chi_s = \left(\frac{\partial m}{\partial h} \right)_{h \rightarrow 0^+}$

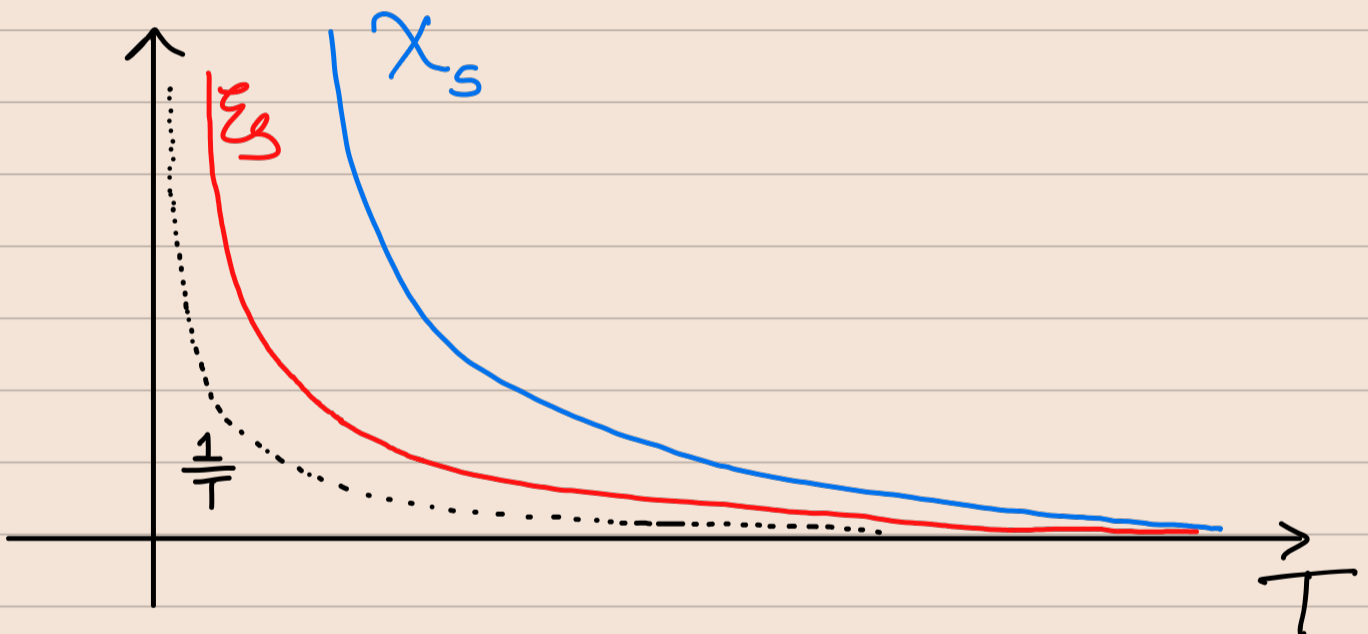
Note: For 1D ising nearest neighbour interaction in the absence of external 'h', we arrived at

$$G(r) = \tanh(\beta J)^r = \langle S_i S_{i+r} \rangle$$

$$\Rightarrow G(r) = e^{-r/\xi_B}, \quad \xi_B = \frac{-1}{\log[\tanh(\beta J)]}$$

($\xi_B > 0$ always)

$$\& \chi_s = B \sum_r G(r) = B \left[\frac{1 + \tanh(\beta J)}{1 - \tanh(\beta J)} \right]$$



As $T \rightarrow 0$, $\xi_B \rightarrow \infty$, Thus

the spins tend to align completely at

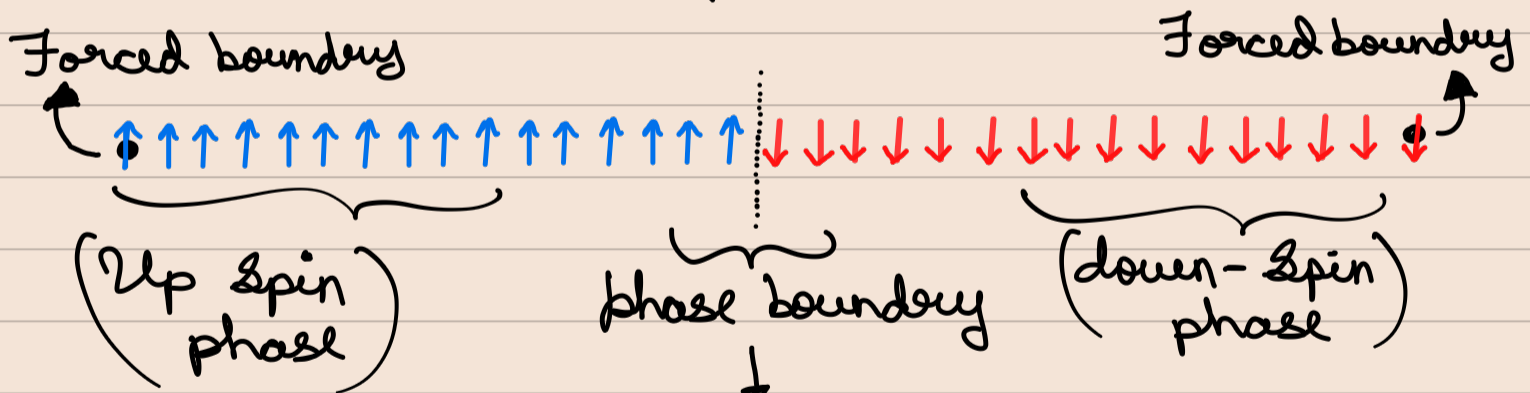
Absolute Zero $\Rightarrow$ Critical temp. T_c is

$T_c = 0 \rightarrow$ for $1 \leq \text{dim.} < 2$ (dim $\in$ fractal)

$T_c > 0 \rightarrow$ for 2D & higher

Anti-Periodic Boundary Conditions

Here we force the Spins on the either end of the chain to be in opposite signs. Thus forcing two different phases (Zero dim. phase boundary) to co-exist at $T=0$.

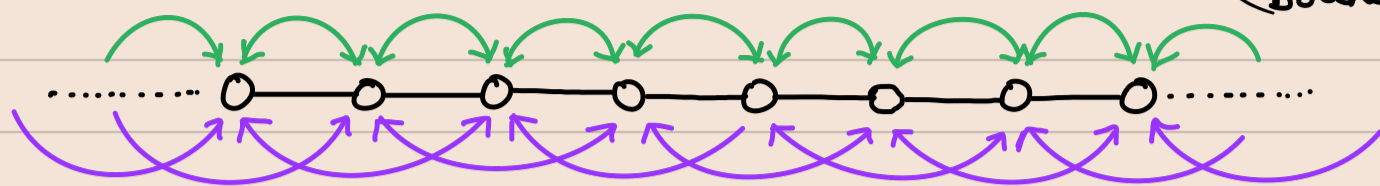


It costs energy due to Surface tension b/w two co-existing phases.

This is how usually Surface tension calculations are performed theoretically.

Further neighbour interactions in 1D Ising Model

(Periodic boundary)



— Further neighbour interaction

— Nearest neighbour interaction

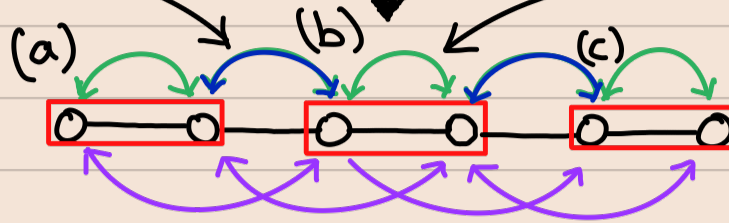
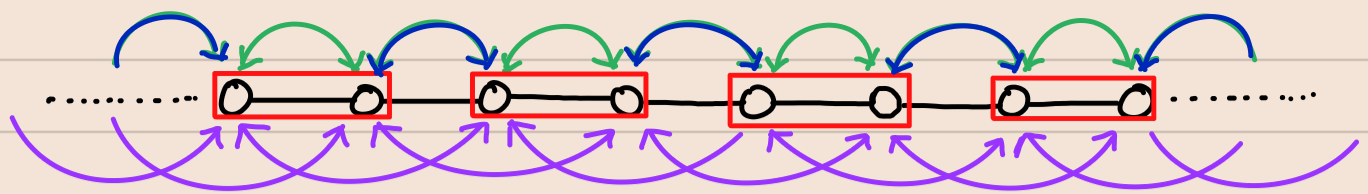
$$\mathcal{H} = -J_1 \sum_{i=1}^N S_i S_{i+1} - J_2 \sum_{j=1}^N S_j S_{j+2}$$

$$\begin{aligned} \hat{\mathcal{H}} = & - \left[(J_1 S_1 S_2 + J_2 S_1 S_3) + (J_1 S_2 S_3 + J_2 S_2 S_4) \right. \\ & + (J_1 S_3 S_4 + J_2 S_3 S_5) + (J_1 S_4 S_5 + J_2 S_4 S_6) \\ & \left. + (J_1 S_5 S_6 + J_2 S_5 S_7) + \dots \right] \end{aligned}$$

$$+ J_1 (S_{N-1} S_N + J_2 S_{N-1} S_1) + (J_1 S_N S_1 + J_2 S_N S_2) \Big]$$

$$\hat{\mathcal{H}} = - \left[\begin{aligned} & J_1 \left(\frac{S_1 S_2 + S_2 S_3 + S_3 S_4}{2} \right) + J_2 (S_1 S_3 + S_2 S_4) \\ & + J_1 \left(\frac{S_3 S_4 + S_4 S_5 + S_5 S_6}{2} \right) + J_2 (S_3 S_5 + S_4 S_6) \\ & + \dots + J_1 (S_{N-1} S_N + S_N S_1 + S_1 S_2) + J_2 (S_{N-1} S_1 + S_N S_2) \end{aligned} \right]$$

Thus the Hamiltonian can be grouped into a sum of two spin containing unit.



This interaction appears twice b/w blocks (a) & (b) and blocks (b) & (c)

— Nearest neighbour interactions that repeats when energy b/w successive blocks are calculated.

— Nearest neighbour interactions that doesn't repeat when energy b/w successive blocks are calculated.

— Further neighbour interactions that doesn't repeat when energy b/w successive blocks are calculated.

Each block has two spins. So, the spin state in a particular block can be represented by,

$$\{|s_i s_{i+1}\rangle\} = \{|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$$

$\langle s_i s_{i+1} | e^{-\beta \hat{H}} | s_{i+2} s_{i+3} \rangle \Rightarrow$ Elements of transfer matrix.

$$[T]_{4 \times 4} = \langle s_1 s_2 | \exp \left(\beta \left[J_1 \left(\frac{s_1 s_2 + s_2 s_3 + s_3 s_4}{2} \right) + J_2 (s_1 s_3 + s_2 s_4) \right] \right) | s_3 s_4 \rangle$$

Thus the partition function can be written as

$$Z = \sum_{\{S_i\}} \langle S_1 S_2 | T | S_3 S_4 \rangle \langle S_3 S_4 | T | S_5 S_6 \rangle \times \dots \langle S_{N-1} S_N | T | S_1 S_2 \rangle$$

$$= \sum_{S_1, S_2} \langle S_1 S_2 | T^{N/2} | S_1 S_2 \rangle$$

$$T = \begin{array}{c} \langle S_1 S_2 | \\ \downarrow \end{array} \begin{array}{c} |S_3 S_4\rangle \rightarrow \\ \begin{array}{cccc} |1,1\rangle & |-1,-1\rangle & |1,-1\rangle & |-1,1\rangle \end{array} \end{array} \begin{bmatrix} \langle 1,1 | & e^{2(K_1+K_2)} & e^{-2K_2} & e^{K_1} & e^{-K_1} \\ \langle -1,-1 | & e^{-2K_2} & e^{2(K_1+K_2)} & e^{-K_1} & e^{K_1} \\ \langle 1,-1 | & e^{-K_1} & e^{K_1} & e^{2(K_1+K_2)} & e^{-2K_2} \\ \langle -1,1 | & e^{K_1} & e^{-K_1} & e^{-2K_2} & e^{2(-K_1+K_2)} \end{bmatrix}$$

Then we can find the largest eigen value λ_0 & then we get

$$\underline{Z \sim \lambda_0^{N/2}}$$

1D Ising chain with $S_i = \{+1, 0, -1\}$

$$\mathcal{H} = -J \sum_{\langle ij \rangle} S_i S_j$$

$$H = \mathcal{H} - hM$$

$$H = -J \sum_i S_i S_{i+1} - h \sum_i S_i$$

$$= - \left[J \sum_i S_i S_{i+1} + \frac{h}{2} \sum_i (S_i + S_{i+1}) \right]$$

We can use transfer matrix method to find Z .

$$Z = \sum_{\{S_i\}} e^{-\beta H} = \sum_{\{S_i\}} e^{\beta \sum_i \left[J S_i S_{i+1} + \frac{h}{2} (S_i + S_{i+1}) \right]}$$

using $\beta J = K$ & $\beta h = L$

$$Z = \sum_{S_1, S_2, \dots, S_N} e^{\left[K S_i S_{i+1} + \frac{L}{2} (S_i + S_{i+1}) \right]}$$

$$Z = \sum_{S_1, S_2, \dots, S_N} \langle S_1 | T | S_2 \rangle \langle S_2 | T | S_3 \rangle \dots \dots \langle S_N | T | S_1 \rangle$$

where $\langle S_i | T | S_{i+1} \rangle$ are elements of T matrix, $S_i, S_{i+1} = \{+1, 0, -1\}$

$$\langle S_i | T | S_{i+1} \rangle = e^{\left[K S_i S_{i+1} + \frac{L}{2} (S_i + S_{i+1}) \right]}$$

$$Z = \sum_{S_1} \langle S_1 | T^N | S_1 \rangle = \text{tr}(T^N)$$

$$T = \begin{bmatrix} e^{K+L} & e^{L/2} & e^{-K} \\ e^{L/2} & 1 & e^{-L/2} \\ e^{-K} & e^{-L/2} & e^{K-L} \end{bmatrix}$$

$$\begin{vmatrix} e^{K+L} - \lambda & e^{L/2} & e^{-K} \\ e^{L/2} & 1 - \lambda & e^{-L/2} \\ e^{-K} & e^{-L/2} & e^{K-L} - \lambda \end{vmatrix} = 0$$

$$(e^{K+L} - \lambda) [(\lambda - 1)(\lambda - e^{K-L}) - e^{-L}]$$

$$- e^{L/2} [e^{L/2} (e^{K-L} - \lambda) - e^{-K} e^{-L/2}]$$

$$+ e^{-K} (1 + [\lambda - 1] e^{-K}) = 0$$

$$\Rightarrow (\lambda - e^{K+L}) [e^{-L} - (\lambda - 1)(\lambda - e^{K-L})] - [e^K - \lambda e^L - e^{-K}] + (\lambda - 1)e^{-2K} + e^{-K} = 0$$

$$\Rightarrow \lambda^3 - (1 + e^{K-L} + e^{K+L})\lambda^2 + [e^{K+L} + e^{K-L} - e^L - e^{-L} + e^{2K} - e^{-2K}]\lambda + e^{-2K} - e^{2K} + 2e^K - 2e^{-K} = 0$$

We get a cubic eqⁿ for ' λ ' as,

$$\Rightarrow \lambda^3 - [1 + 2e^K \cosh(L)]\lambda^2 + [2(e^K - 1)\cosh(L) + 2\cosh(2K)]\lambda + 4\sinh(K) - 2\sinh(2K) = 0$$

In the absence of magnetic field we get eigen values as a root

of

$$\lambda^3 - (1 + 2e^K)\lambda^2 + 2[(e^K - 1) + \cosh(2K)]\lambda + 4\sinh(K) - 2\sinh(2K) = 0$$

$$\Rightarrow [(\lambda + 1)(\lambda - 2) - 2(\lambda - 1)\cosh(K)][\lambda - 2\sinh(K)] = 0$$

$$\Rightarrow \lambda = 2\sinh(K) \quad \text{or}$$

$$\lambda^2 - [1 + 2\cosh(K)]\lambda - 2[1 - \cosh(K)] = 0$$

$$\Rightarrow \lambda = \frac{[1 + 2\cosh(K)] \pm \sqrt{[1 + 2\cosh(K)]^2 + 8[1 - \cosh(K)]}}{2}$$

$$\Rightarrow \lambda_0 = \frac{1 + 2\cosh(K) + \sqrt{2\cosh(2K) - 4\cosh(K) + 1}}{2}$$

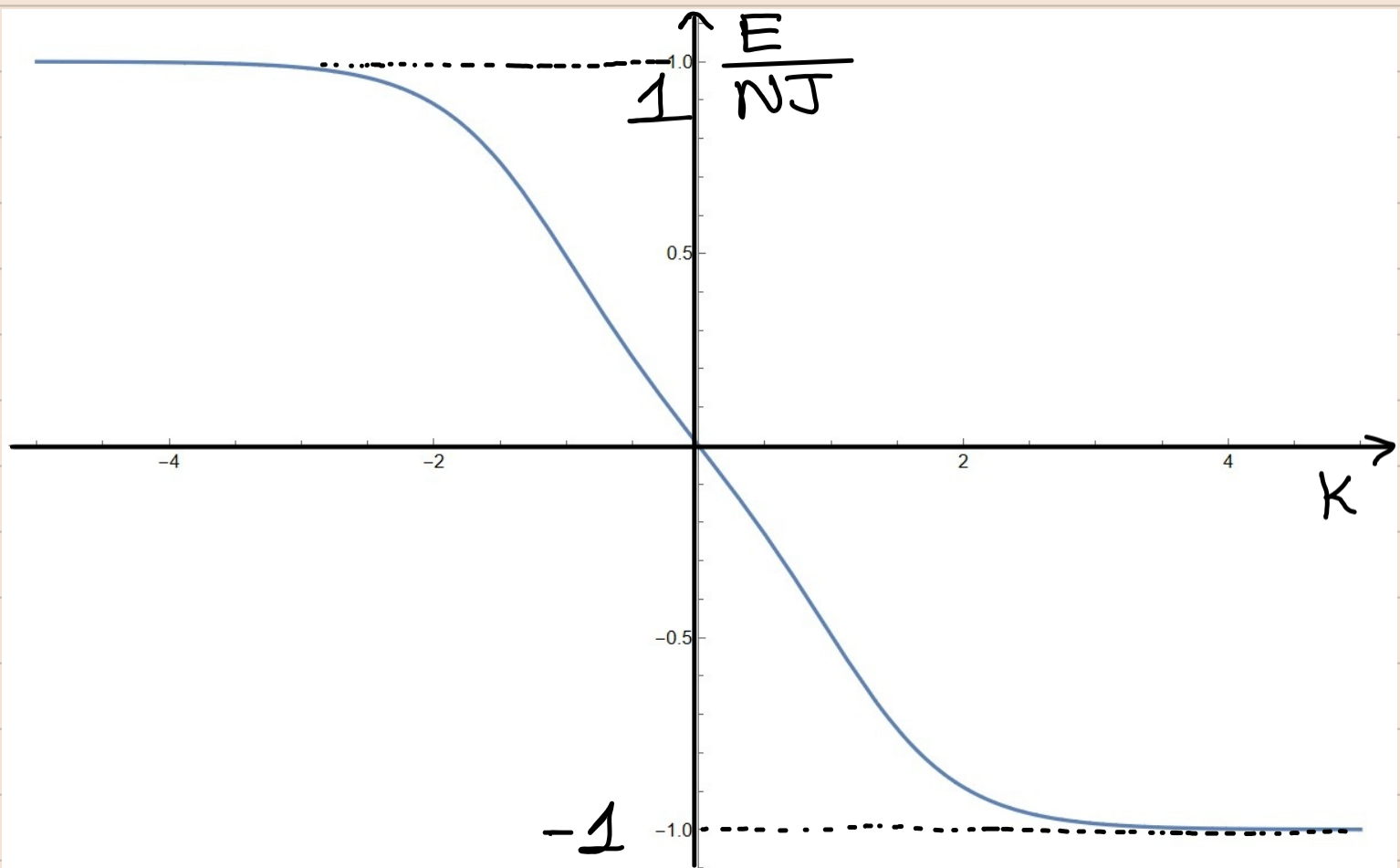
$$\lambda_1 = 2\sinh(K)$$

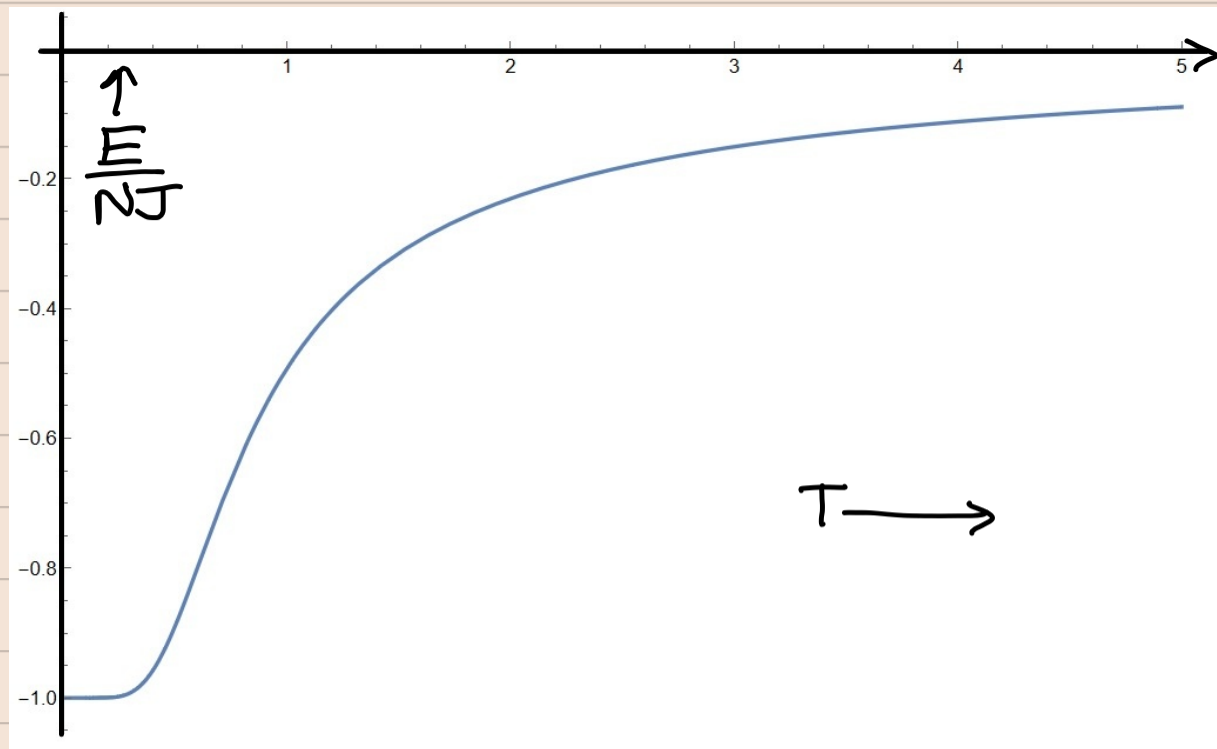
$$\lambda_2 = \frac{1 + 2\cosh(K) - \sqrt{2\cosh(2K) - 4\cosh(K) + 1}}{2}$$

So λ_0 is the eigen value whose magnitude is largest for all K .

$$\Rightarrow Z \sim \lambda_0^N \Rightarrow E = -\frac{\partial}{\partial \beta} \ln(Z)$$

$$\Rightarrow \frac{E}{J} = -\frac{\partial \ln(Z)}{\partial K} \Rightarrow \frac{E}{NJ} = -\frac{\partial \ln(\lambda_0)}{\partial K}$$

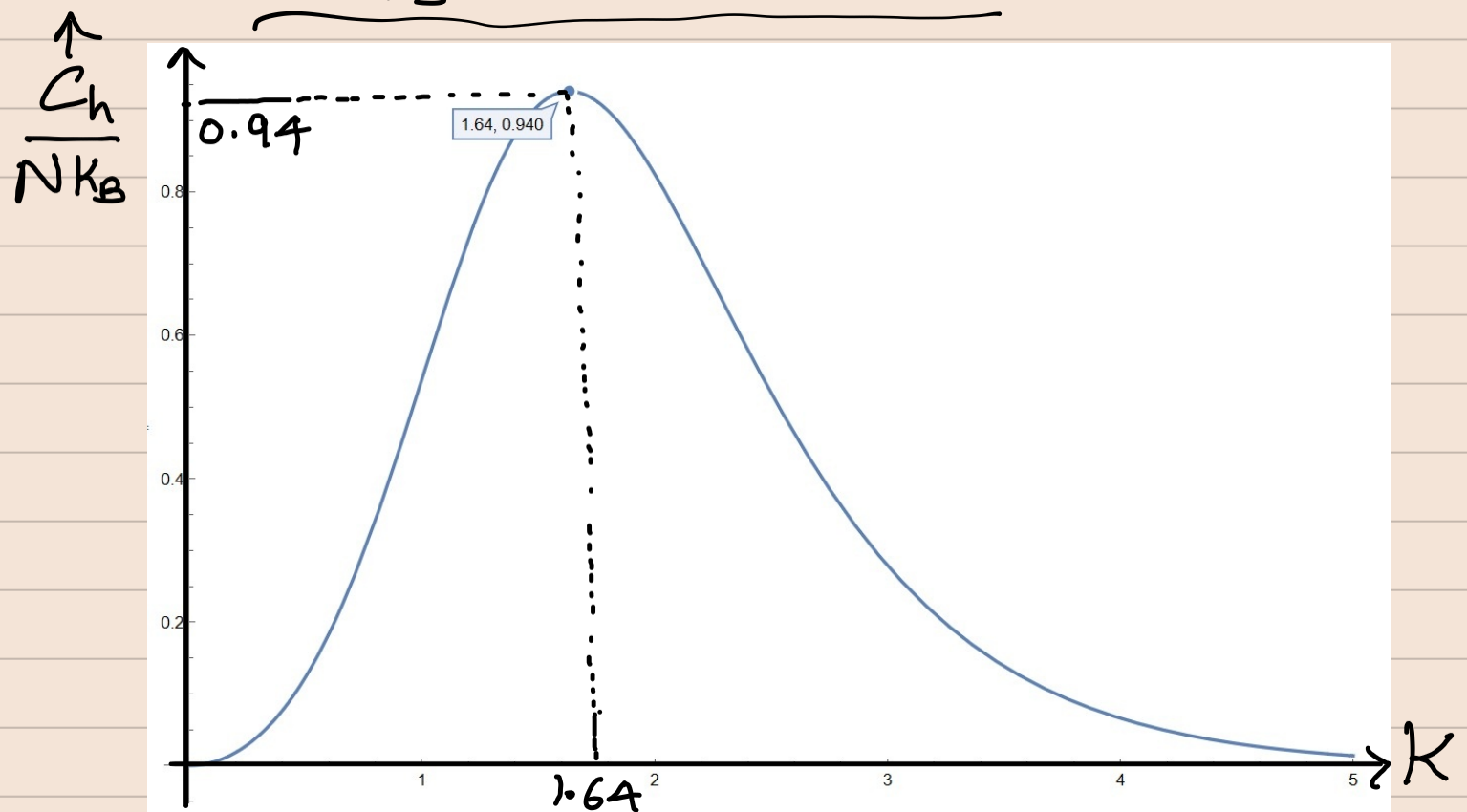




$$C_h = \frac{\partial E}{\partial T} = \frac{\partial \beta}{\partial T} \left(\frac{\partial E}{\partial \beta} \right) = k_B \beta^2 \frac{\partial^2}{\partial \beta^2} \ln(Z)$$

$$\Rightarrow C_h = k_B K^2 \frac{\partial^2}{\partial K^2} \ln(Z) = N k_B K^2 \left[\frac{\partial^2}{\partial K^2} \ln(\lambda_0) \right]$$

$$\Rightarrow \frac{C_h}{N k_B} = K^2 \frac{\partial^2}{\partial K^2} \ln(\lambda_0)$$



Potts Model

1D & nearest neighbour interactions



Here each spin variable can take values of $S_i = \{1, 2, 3, \dots, q\}$ ($q \geq 2$)

The hamiltonian is given by

$$\underline{H_p = -J_p \sum_{i,j} \delta(S_i, S_j)}$$

for nearest neighbour interactions we

get $H_p = -J_p \sum_i \delta(S_i, S_{i+1})$

$$Z = \sum_{\{S_i\}} e^{\beta J_p \sum_i \delta(S_i, S_{i+1})}$$

$$Z = \sum_{S_1, S_2, \dots, S_N} e^{\beta J_p \delta(S_1, S_2)} e^{\beta J_p \delta(S_2, S_3)} \dots \times e^{\beta J_p \delta(S_N, S_1)}$$

$$Z = \sum_{S_1, S_2, \dots, S_N} \langle S_1 | T | S_2 \rangle \langle S_2 | T | S_3 \rangle \times \dots \times \dots \times \langle S_N | T | S_1 \rangle$$

$$\Rightarrow Z = \sum_{S_1} \langle S_1 | T^N | S_1 \rangle = \text{tr}(T^N)$$

$$T = \begin{bmatrix} e^k & 1 & 1 & \dots & 1 \\ 1 & e^k & 1 & \dots & 1 \\ 1 & 1 & e^k & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & e^k \end{bmatrix}_{q \times q}$$

$$\begin{bmatrix} e^k & 1 & 1 & \dots & 1 \\ 1 & e^k & 1 & \dots & 1 \\ 1 & 1 & e^k & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & e^k \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \frac{1}{\sqrt{q}} = (e^k + q - 1) \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \frac{1}{\sqrt{q}}$$

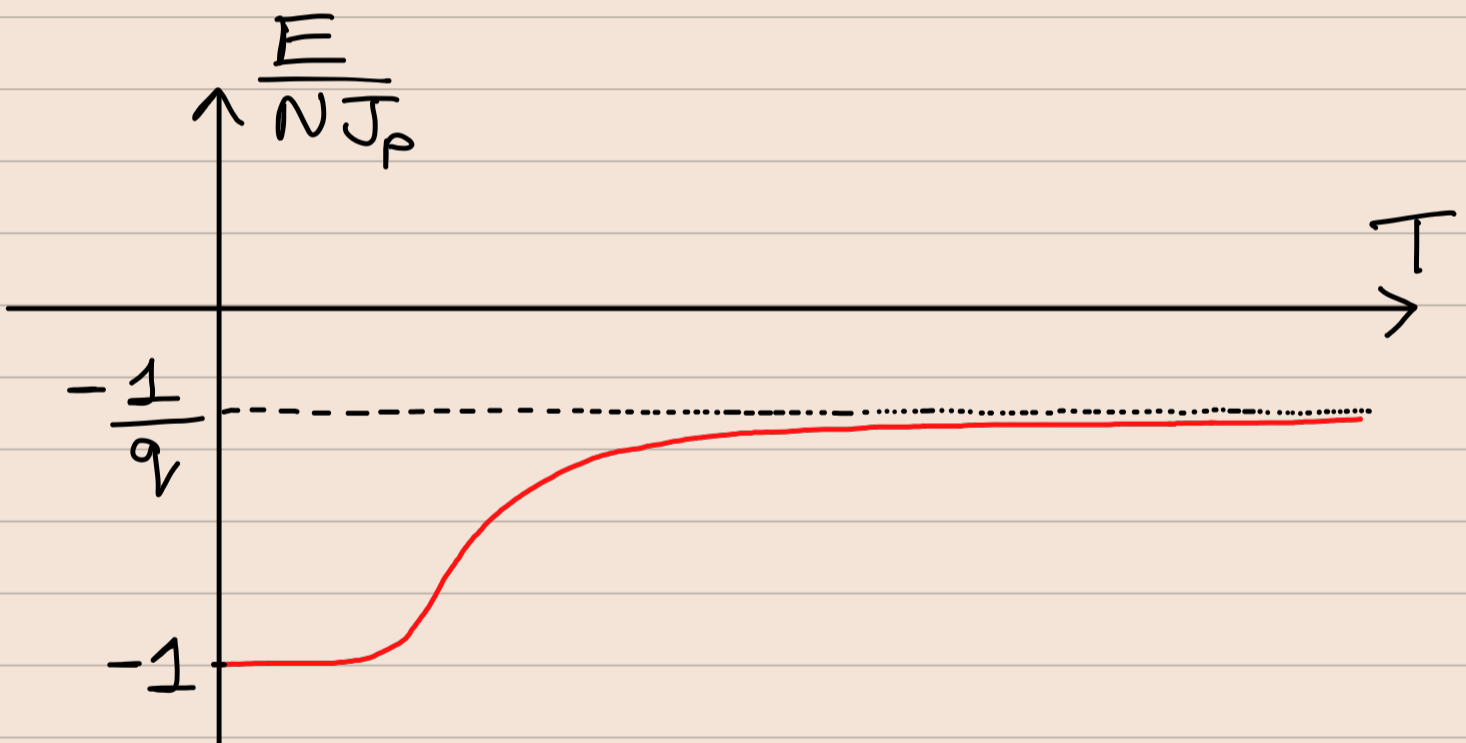
Thus the largest eigen value is given by $\lambda_0 = e^k + q - 1$ with the eigen vector $\frac{1}{\sqrt{q}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{q \times 1}$

$$\Rightarrow \underline{Z = (e^K + q - 1)^N}$$

$$E = - \frac{\partial}{\partial \beta} \ln(Z) = -NJ_p \frac{\partial}{\partial K} \ln(e^K + q - 1)$$

$$E = -NJ_p \left[\frac{e^K}{e^K + q - 1} \right]$$

$$E = -NJ_p \frac{e^{\beta J_p}}{(e^{\beta J_p} + q - 1)}$$



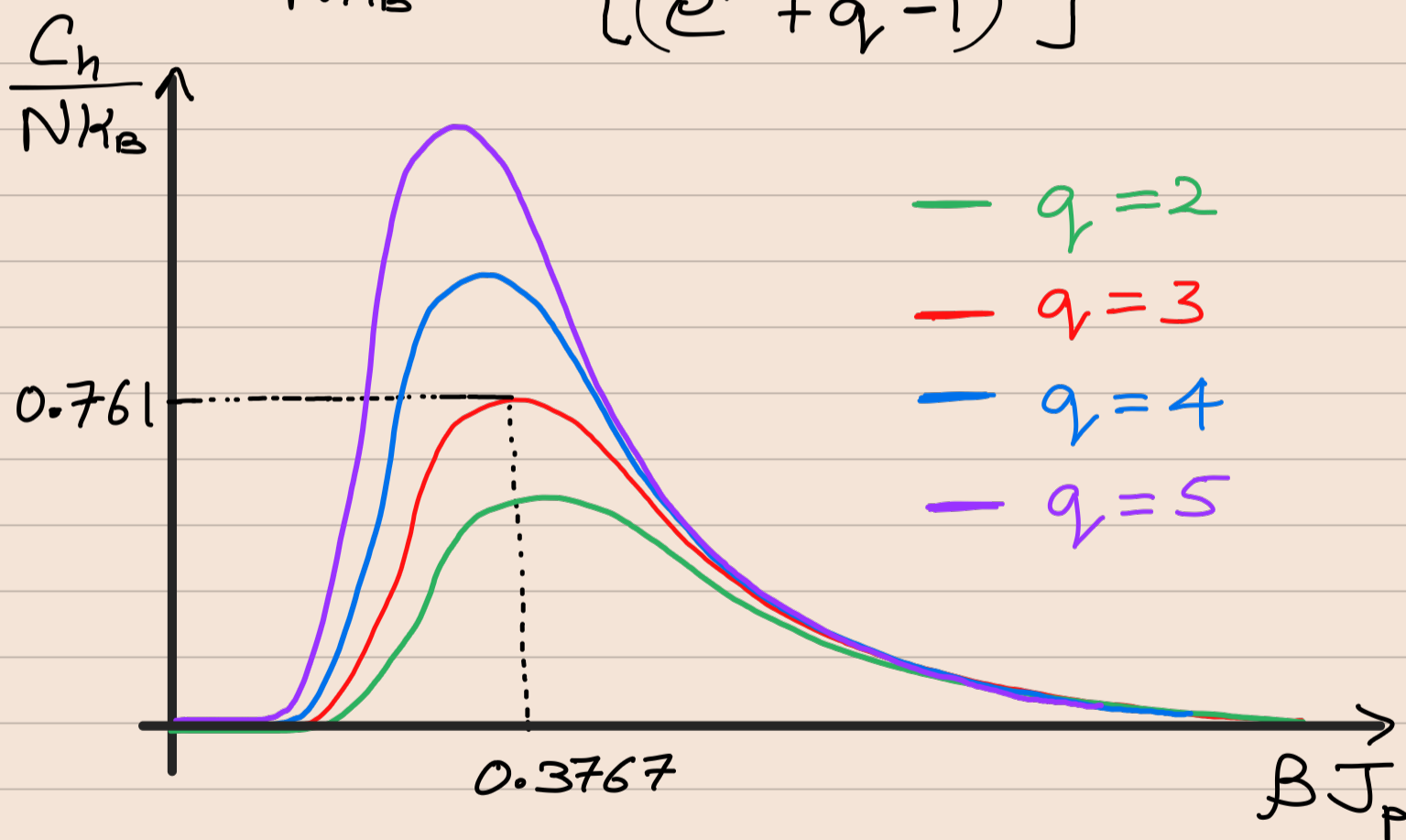
$$C_h = \frac{\partial E}{\partial T} = K_B \beta^2 \frac{\partial^2}{\partial \beta^2} \ln(Z)$$

$$\Rightarrow \underline{\frac{C_h}{NK_B} = K^2 \frac{\partial^2}{\partial K^2} \ln(\lambda_0)}$$

$$\frac{C_h}{Nk_B} = k^2 \frac{\partial}{\partial k} \left[\frac{e^k}{(e^k + q - 1)} \right]$$

$$= k^2 \frac{(e^k + q - 1)e^k - e^{2k}}{(e^k + q - 1)^2}$$

$$\frac{C_h}{Nk_B} = k^2 \left[\frac{e^k (q - 1)}{(e^k + q - 1)^2} \right]$$



Note: 2D potts model with $q=2$ is equivalent to 2D ising model.

And for $q > 2$ the 2D potts model is still unsolved.

Equivalence of 1D Ising model &

1D Potts model with $q=2$

1D Ising model

$$\mathcal{H}_I = -J_I \sum_i \vec{S}_i \cdot \vec{S}_{i+1}$$

$$\vec{S}_i = \{+1, -1\}$$

For $i=1$,

$$\mathcal{H}_I(1,1) = -J_I$$

$$\mathcal{H}_I(1,-1) = +J_I$$

$$\mathcal{H}_I(-1,1) = +J_I$$

$$\mathcal{H}_I(-1,-1) = -J_I$$

1D Potts model ($q=2$)

$$\mathcal{H}_P = -J_P \sum_i \delta(S_i, S_{i+1})$$

$$S_i = \{1, 2\}$$

For $i=1$,

$$\mathcal{H}_P(2,2) = -J_P$$

$$\mathcal{H}_P(2,1) = 0$$

$$\mathcal{H}_P(1,2) = 0$$

$$\mathcal{H}_P(1,1) = -J_P$$

Consider $\mathcal{H}_I = A \mathcal{H}_P + B$

$$\Rightarrow -J_I = A(-J_P) + B$$

$$\& J_I = A(0) + B$$

$$\Rightarrow B = J_I \quad \& \quad A = \frac{2J_I}{J_P}$$

$$\Rightarrow \boxed{\mathcal{H}_I = \frac{2J_I}{J_P} \mathcal{H}_P + J_I}$$